



# **Data Weaver - Distributed Agentic Data Pipeline - Capstone Project**

**Team - 2**

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# Team Members & Roles



**Tenzing Salaka**

AI Engineer - Engineer Agent



**Shreya Anand Kuvalekar**

Scrum Master/UI Developer/System Architect



**Orges Velia**

AI Engineer-Architect Agent





**Priya Prajapati**

Backend & Data Engineer



**Kushwanth Reddy Nomula**

Frontend / Dashboard Engineer



**Meghana Ventrapragada**

Backend & Data Engineer

# Problem Statement

Organizations accumulate fragmented, inconsistent, and poorly documented datasets across departments, resulting in “data swamp” conditions that hinder analytics and decision-making. Integrating such datasets requires significant manual effort to infer schema relationships, identify primary and foreign keys, resolve type mismatches, and validate joins. Existing LLM-based data tools attempt automation but rely on probabilistic reasoning without deterministic safeguards, often hallucinating schema mappings or producing unsafe joins that corrupt downstream analytics. Additionally, most solutions lack distributed scalability and structured separation between semantic reasoning and execution logic. There is a need for a reliable, distributed, self-correcting AI system that separates semantic modeling from deterministic data execution while enforcing invariant-based validation to ensure trustworthy data integration.

# Project Description

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Name:        | Data Weave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Team:                | Team 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Project Description: | <p>For data analysts, data engineers, and organizations dealing with fragmented datasets who struggle to clean, integrate, and understand inconsistent and unstructured data sources the Distributed Agentic Data Pipeline is a multi-agent, AI-driven data integration system using a distributed architecture that automatically maps, cleans, validates, and merges datasets while generating reliable data models and documentation</p> <p>Unlike traditional ETL tools or LLM-based solutions that rely on manual configuration or produce unreliable, unverifiable outputs</p> <p>Our application uses a self-correcting two-agent architecture with deterministic validation checks to ensure accuracy, scalability, and trustworthy data transformation</p> |
|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Benefit Outcomes:    | <ul style="list-style-type: none"><li>• Better data integration decisions with AI-assisted schema inference and validation checks</li><li>• Reduced ETL troubleshooting time through automated mapping, profiling, and transformation previews</li><li>• Improved trust through transparent logs, ERD visibility, and human approval checkpoints</li><li>• Faster delivery of analytics-ready datasets and data dictionaries for downstream reporting</li></ul>                                                                                                                                                                                                                                                                                                     |
| GitHub Link:         | <a href="https://github.com/htmw/S2026A-Team2">https://github.com/htmw/S2026A-Team2</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

# Team Working Agreement

## Team Principles

- Kindness, respect, and patience in all interactions
- Clear ownership and responsibilities
- Even workload distribution; break large tasks
- Document attempted solutions before escalating
- Use GitHub for all updates and reviews

## Checklist to Mark Task Complete

- Code runs with no syntax/runtime errors
- Meets acceptance criteria and tests
- Peer review and multi-member verification
- Push to branch and open PR with docs updated
- All relevant documentation tagged and reviewed





**Alex Martinez — Senior Data Engineer**

**Industry: E-commerce / Healthcare / FinTech**

**Experience: 5–8 years**

## Description

It is designed to streamline the integration of fragmented legacy data sources by automating schema interpretation and relationship mapping. It introduces deterministic validation checks—such as row-count reconciliation and null-threshold monitoring—to ensure transformation accuracy. The platform also offers end-to-end transparency into data processing steps and produces downloadable, analysis-ready datasets along with comprehensive data dictionaries.

## Goals

- Consolidate multiple unstructured CSV exports originating from legacy systems
- Enable safe, accurate joins while maintaining schema consistency
- Minimize time spent on manual ETL troubleshooting
- Provide dependable, analytics-ready datasets to downstream teams

## Challenges

Non-standardized column naming conventions (e.g., *cust\_id*, *customerID*, *user\_ref*)

- Lack of documentation for legacy data structures
- Join amplification leading to inflated or duplicated row counts
- Undetected schema mismatches corrupting downstream dashboards
- Excessive time spent validating transformations rather than developing pipelines



🗒️ **Priya Sharma — BI Analyst**  
**Industry: Retail / SaaS**  
**Experience: 3–5 years**

### **Description**

This system empowers analytics and reporting teams to independently prepare and understand integrated datasets. It provides visual entity-relationship diagrams (ERDs) to clarify how tables connect, along with transparent pipeline logs that document validation steps. Users can download analysis-ready datasets and leverage auto-generated data dictionaries to better interpret column meanings—ensuring confidence in every output produced.

### **Goals**

- Rapidly merge datasets to support reporting needs
- Gain clear visibility into table relationships through ERD views
- Access clean, dependable data without relying on engineering teams
- Build dashboards powered by accurate, trustworthy metrics

### **Challenges**

- Delays caused by dependence on data engineering for cleaned datasets
- Ambiguity around primary and foreign key columns
- Duplicate records distorting KPI and metric calculations
- Insufficient metadata and lack of formal data dictionaries
- Skepticism toward AI-generated joins without validation guarantees



 **Daniel Kim — Chief Technology Officer / Head of Data**

**Industry: SaaS / FinTech / Healthcare**

**Experience: 10+ years**

### **Description**

This system is built to deliver enterprise-grade data reliability by combining deterministic validation with scalable AI infrastructure. It introduces invariant-based checks to ensure transformation accuracy, while maintaining full audit trails for governance and compliance. The architecture is designed for distributed, production-ready deployment and automatically generates documentation—including ERDs and data dictionaries—so organizations can trust, verify, and operationalize AI-driven data outputs at scale.

### **Goals**

- Maintain consistent, high-quality data across all departments
- Lower engineering effort required for complex data integrations
- Strengthen governance, traceability, and audit readiness
- Implement scalable, AI-enabled infrastructure
- Showcase innovation through distributed AI system adoption

### **Challenges**

- Data inconsistencies leading to conflicting reports across departments
- Significant engineering costs tied to building and maintaining ETL pipelines
- Limited schema transparency and insufficient documentation
- Concerns around AI systems generating unverified or non-auditable outputs
- Legacy or single-node infrastructure that cannot scale with enterprise demands



# Minimum Viable Product

- ❖ Ingestion & Scout: Upload CSV/API via Stream lit UI; profiling, schema sniffing, PK/FK hints.
- ❖ Architect: Schema inference, ERD creation, data dictionary generation.
- ❖ Engineer: Generates Polars/Pandas code, cleaning steps, validation rules.
- ❖ Router + LLM Retry Loop: Auto-repair failed runs, regenerate improved transformations.
- ❖ Invariant Enforcement: Deterministic checks; halt + logs on violation.
- ❖ Loader & Storage: Successful pipeline saved to SQLite with logs + lineage.
- ❖ Dashboard: Stream lit UI shows ERD, profiling, invariants, logs, downloads.



# Technologies & Algorithms

- **Architecture & Orchestration:** LangGraph — multi-agent cycles with explicit checkpoints
- API Layer: FastAPI for inter-node communication and control



- Networking: Tailscale for secure distributed connectivity

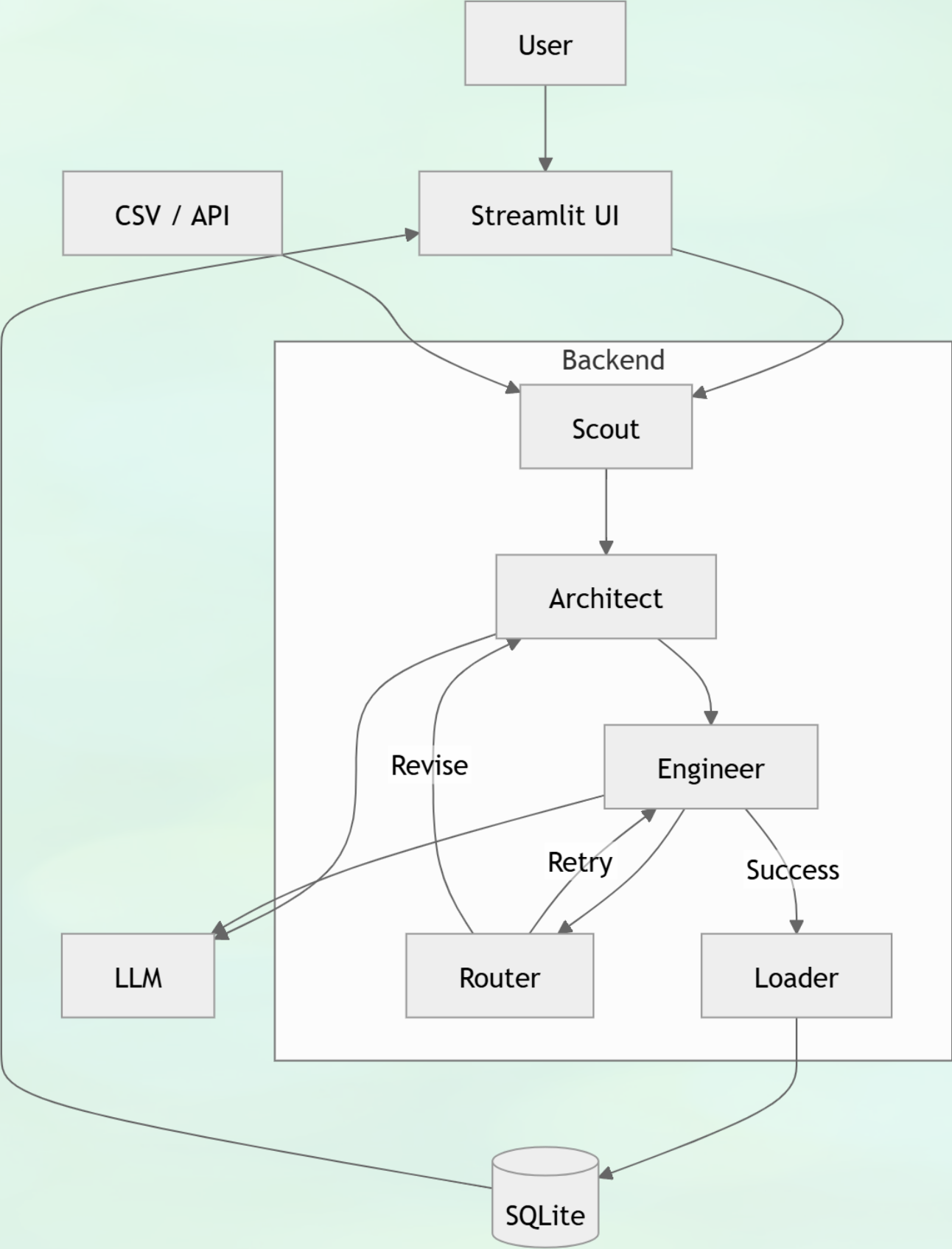


- UI: Streamlit dashboard for visualization & human checkpoints
- **Models:** local gpt-oss:20b (quantized) — Mapper high-reasoning, Aggregator high-throughput



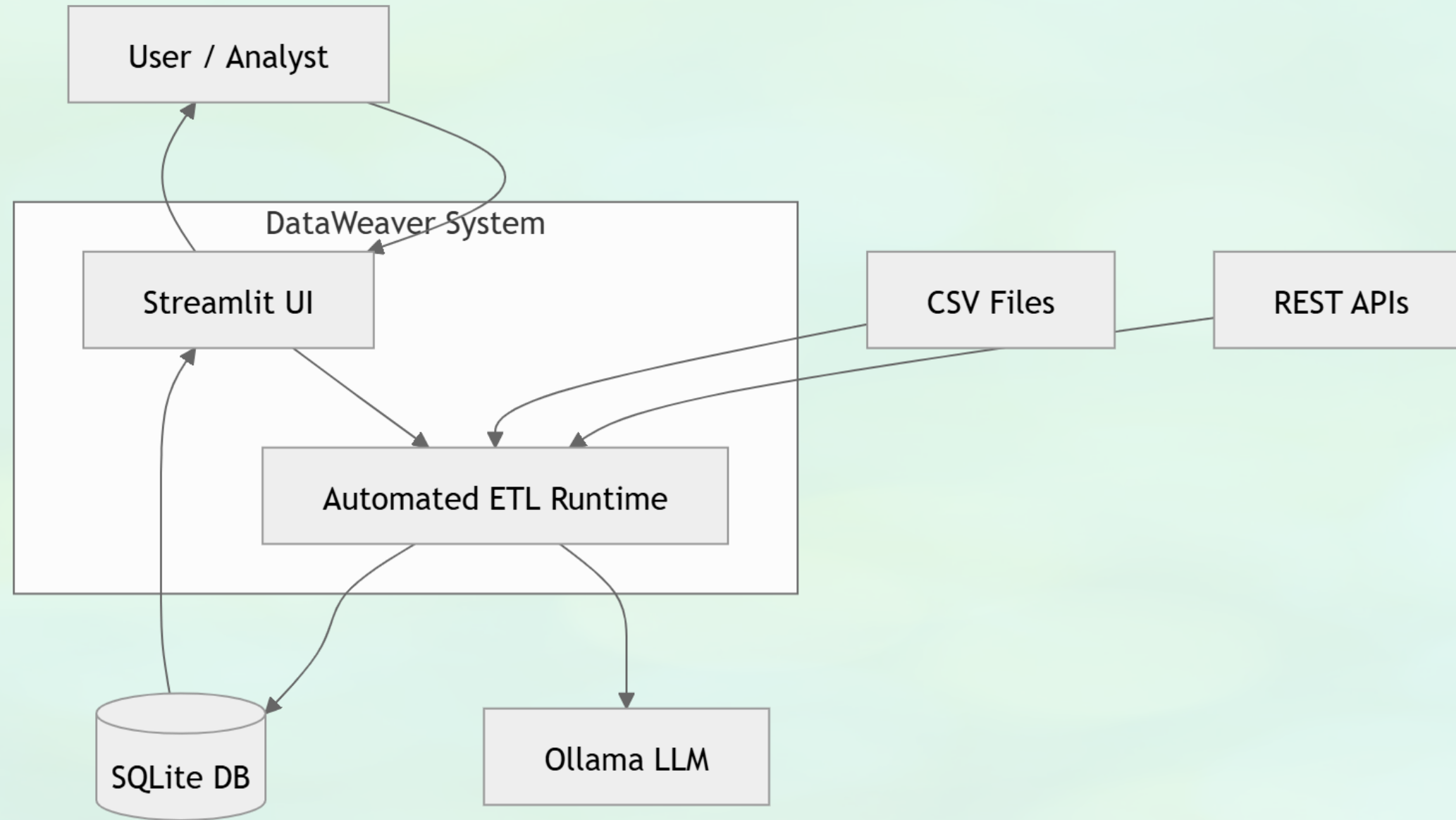
- **Data Processing:** Polars / Pandas (GPU-accelerated when available)

# Architecture Diagram

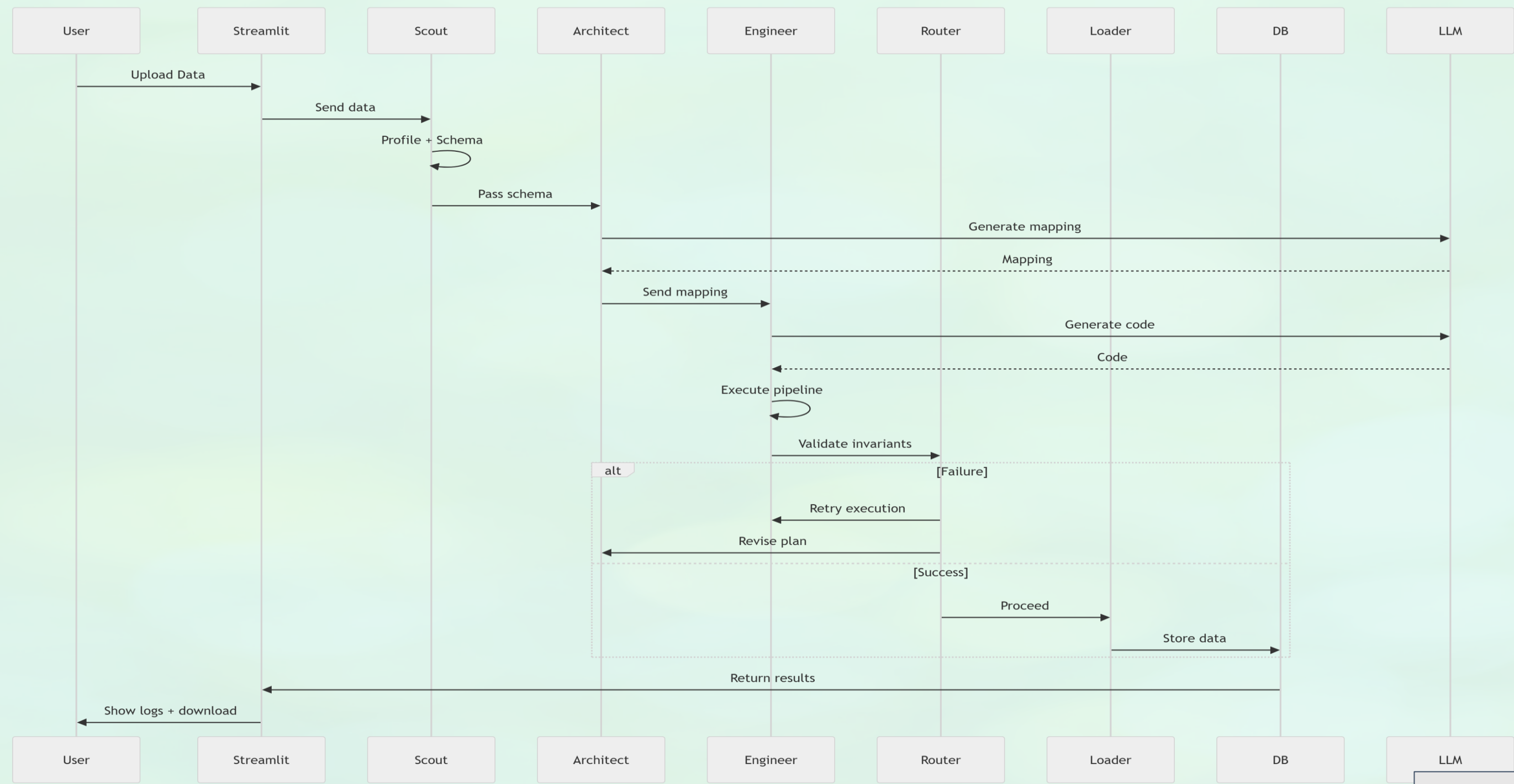




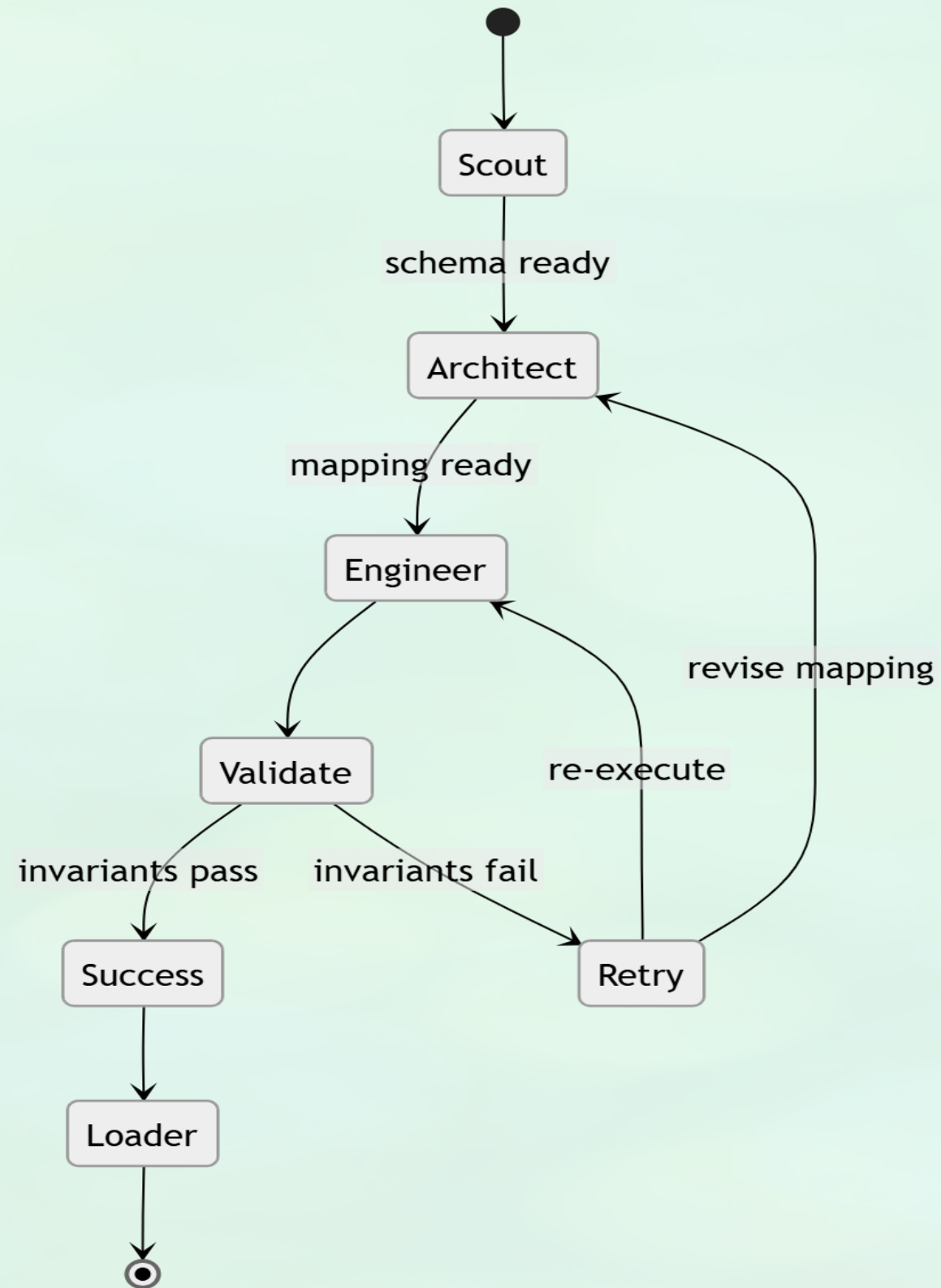
# Context diagram



# Sequence Diagram

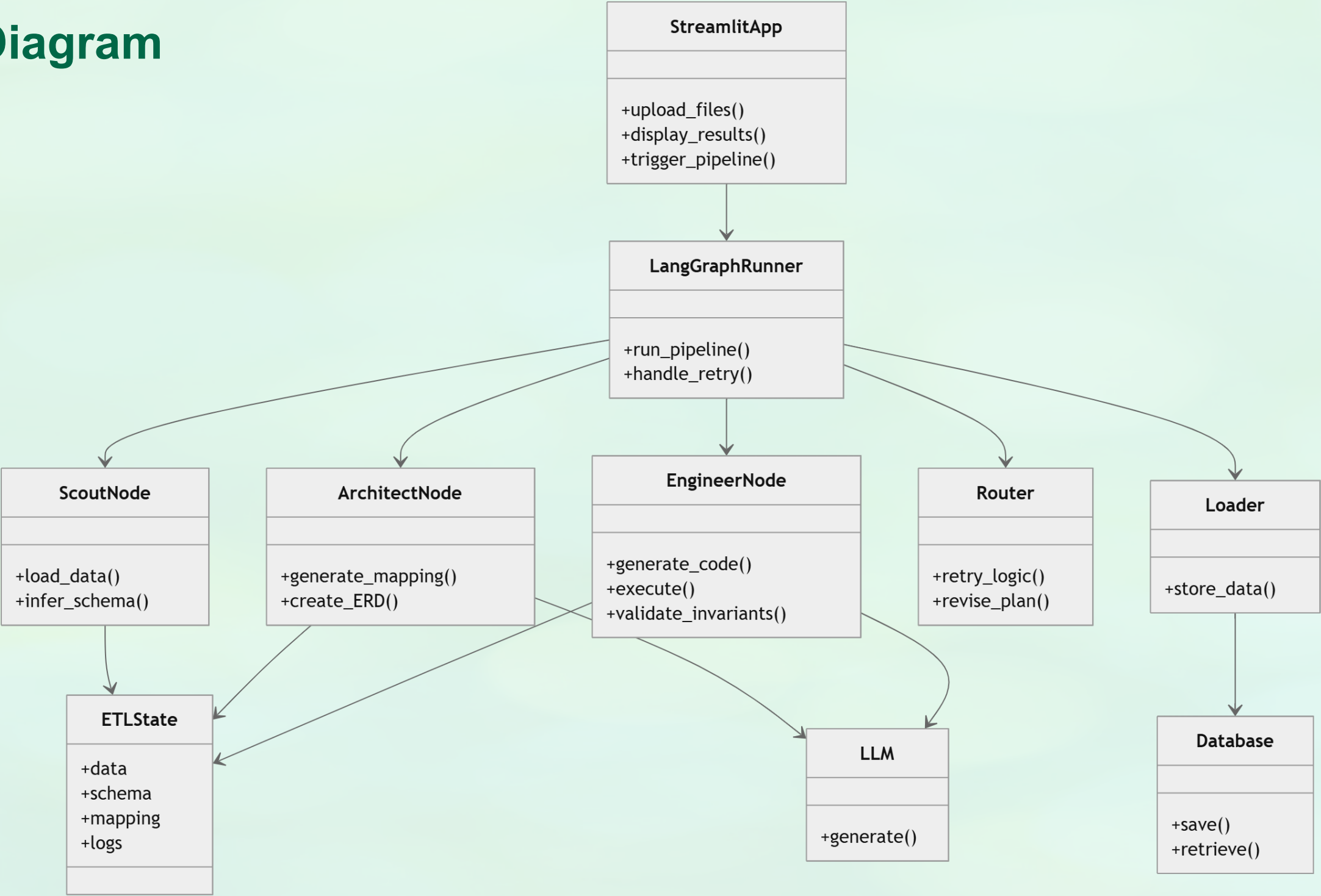


## State Diagram





# Class Diagram



# Sprint 2 - Recap

The primary objective of Sprint 2 was to improve the robustness of the distributed agentic pipeline through advanced validation checks, monitoring systems, and enhanced export/documentation capabilities.

## Major Features Completed

- **US\_006 – Row Count Validation**

Implemented invariant-based row count validation to prevent unsafe joins and detect data amplification issues during transformations. This ensured that merged datasets maintained logical consistency and reduced the risk of duplicated or inflated records.

- **US\_007 – Schema Validation**

Developed schema verification mechanisms to confirm that transformed datasets matched expected target schemas. This improved downstream compatibility and reduced errors caused by datatype mismatches.

- **US\_008 – Null Detection**

Integrated null-threshold monitoring to identify spikes in missing values across transformed datasets. The feature enabled early detection of data quality degradation and improved pipeline reliability.

# Sprint 2 - Recap

- **US\_009 – Pipeline Monitoring**

Built real-time monitoring and execution tracking capabilities for the distributed pipeline. Users could now visualize processing stages, validation checkpoints, and execution logs through the Streamlit dashboard.

- **US\_010 – Data Download**

Added downloadable output functionality, allowing users to export cleaned and transformed datasets directly from the platform for analytics and reporting purposes.

## **Key Outcomes**

- Improved trustworthiness of AI-driven data transformations
- Increased transparency through monitoring and validation logs
- Reduced risk of schema inconsistencies and incorrect joins
- Enhanced user accessibility with downloadable outputs
- Strengthened foundation for distributed processing capabilities planned in Sprint 3



# Product Backlog

| 1  | ID    | Assigned      | Name                             | Type | Creation Date | US                                                                                                                                                                                                              | Size   | Priority | Dependencies      |
|----|-------|---------------|----------------------------------|------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|-------------------|
| 2  | US1   | Orges/Tenzing | API & CSV Tooling                | F    | 2025-03-11    | As a data engineer, I want the agent to have tools to connect to APIs and read CSV files, so that I can start extraction.                                                                                       | Small  | High     | None              |
| 3  | US2   | Orges/Tenzing | Data Extraction & Transformation | F    | 2025-03-11    | As a data engineer, I want the agent to extract and transform data from APIs and CSVs, so I can integrate sources.                                                                                              | Small  | High     | US1               |
| 4  | US3   | Orges/Tenzing | Sensitive Data Redaction         | F    | 2025-03-11    | As a compliance officer, I want the agent to redact sensitive data, so we ensure privacy.                                                                                                                       | Medium | Medium   | US2               |
| 5  | US4   | Orges/Tenzing | ETL Step Verification            | F    | 2025-03-11    | As a data steward, I want to verify each ETL step, so we ensure quality.                                                                                                                                        | Small  | High     | US2               |
| 6  | US5   | Orges/Tenzing | Agent Coordination Framework     | F    | 2025-03-11    | As a team lead, I want a coordination framework, so agents collaborate on tasks.                                                                                                                                | Medium | Medium   | None              |
| 7  | US6   | Orges/Tenzing | Parallel Multi-Agent Execution   | F    | 2025-03-11    | As a team lead, I want multiple agents to collaborate in parallel, so complex tasks finish faster.                                                                                                              | Large  | Medium   | US5               |
| 8  | US7   | Shreya/Kush   | Schema Change Monitoring         | F    | 2025-03-11    | As a data steward, I want to monitor dataset headers, so we catch schema changes.                                                                                                                               | Medium | Medium   | US2               |
| 9  | US8   | Shreya/Kush   | Pre-Join Data Sampling           | F    | 2025-03-11    | As a product owner, I want the agent to sample data before joining, so only clean data is used.                                                                                                                 | Medium | Medium   | US2               |
| 10 | US9   | Orges/Tenzing | Conversational Memory            | F    | 2025-03-11    | As a user, I want the agent to remember past requests, so interactions are contextual.                                                                                                                          | Large  | Low      | Later enhancement |
| 11 | US9.1 | Priya/Meghana | Pipeline Performance             | NF   | 2025-03-11    | As a data engineer, I want the ETL pipeline to complete a standard extraction and transformation job within a reasonable time threshold, so that the system is practical for real workloads and not just demos. | Medium | Medium   | US1               |

# Product Backlog

|    |      |               |                                                  |    |            |                                                                                                                                                                                                   |        |        |                                          |
|----|------|---------------|--------------------------------------------------|----|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|------------------------------------------|
| 12 | US10 | Priya/Meghana | Fault Tolerance                                  | NF | 2025-03-11 | As a data engineer, I want the pipeline to gracefully handle agent failures without crashing the entire run, so that a single broken step doesn't require restarting the whole job from scratch.  | Medium | High   | US1                                      |
| 13 | US11 | ongoing       | Credential & API Key Security                    | NF | 2025-03-11 | As a compliance officer, I want API keys and database credentials to be stored securely and never logged in plain text, so that sensitive configuration data is not exposed in traces or outputs. | Small  | High   | US1                                      |
| 14 | US12 | ongoing       | Model Resource Allocation                        | NF | 2025-03-11 | As a data engineer, I want LLM-based agents to be assigned to the appropriate GPU based on model size and VRAM requirements, so that the pipeline runs efficiently without memory errors.         | Small  | High   | None                                     |
| 15 | US13 | Priya/Meghana | Pipeline Maintainability                         | NF | 2025-03-11 | As a developer, I want each agent to be modular and independently testable, so that changes to one agent don't require refactoring the entire pipeline.                                           | Medium | Medium | None                                     |
| 16 | US14 | ongoing       | Traceability & Auditability                      | NF | 2025-03-11 | As a data steward, I want every agent action and decision to be logged with a timestamp and input/output summary, so that the pipeline can be audited after a run completes.                      | Small  | High   | US6                                      |
| 17 | US15 | sprint 3      | Scalability                                      | NF | 2025-03-11 | As a data engineer, I want the pipeline to handle datasets of increasing size without significant performance degradation, so that it remains viable beyond small demo datasets.                  | Large  | Low    | Later enhancement                        |
| 18 | US16 | Shreya/Kush   | Error Detection During Extraction/Transformation | F  | 2026-03-11 | As a user, I want the system to detect and notify errors during data extraction or transformation so that I can correct issues before processing continues.                                       | Medium | High   | Data processing engine, validation rules |

# Product Backlog

|    |      |             |                                            |    |            |                                                                                                                                                                                                                                                                                                                                                  |        |        |                                                       |
|----|------|-------------|--------------------------------------------|----|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-------------------------------------------------------|
| 19 | US17 | Shreya/Kush | Multi-Format Output Export                 | F  | 2026-03-11 | As a user, I want to export processed data in formats like CSV, Excel, Google Sheets, and JSON so that I can use the data in other tools.                                                                                                                                                                                                        | Medium | High   | File export service, Google Sheets API                |
| 20 | US18 | Shreya/Kush | Drag & Drop / API / Link Upload            | F  | 2026-03-11 | As a user, I want to upload files via drag & drop or provide API links in a textbox so that I can easily ingest data from different sources.                                                                                                                                                                                                     | Large  | High   | uploader, API connector service                       |
| 21 | US19 | Shreya/Kush | Database Preview Before & After Processing | F  | 2026-03-11 | As a user, I want to preview the database before and after processing in table or ERD format so that I can understand the structure and transformations applied.                                                                                                                                                                                 | Large  | Medium | Database visualization library, transformation engine |
| 22 | US20 | Shreya/Kush | Basic Analytics Functions                  | F  | 2026-03-11 | As a user, I want basic analytics like average, median, and summary stats so that I can quickly analyze my dataset without external tools.                                                                                                                                                                                                       | Medium | Medium | Data processing engine, aggregation functions         |
| 23 | US21 | Shreya/Kush | Dropdown Filters for Data                  | F  | 2026-03-11 | As a user, I want dropdown filtering options so that I can easily filter datasets and focus on relevant information.                                                                                                                                                                                                                             | Small  | Medium | UI filtering components, dataset query layer          |
| 24 | US22 | ongoing     | Data Quality Validation                    | F  | 2026-03-11 | As a user, I want the system to validate dataset quality so that I can detect missing values, duplicates, and invalid data before processing.<br><b>Acceptance Criteria</b><br>Detect missing values<br>Detect duplicate rows<br>Validate column data types<br>Generate validation summary                                                       | Medium | Medium |                                                       |
| 25 | US23 | sprint 3    | Dataset Health Score                       | NF | 2026-03-11 | As a user, I want the system to generate a dataset health score so that I can quickly understand the overall quality of my dataset.<br><b>Acceptance Criteria</b><br>Health score calculated from validation metrics<br>Score range from 0-100<br>Health classification displayed (Healthy, Moderate, Poor)<br>Score displayed in dataset report | Small  | Medium |                                                       |



# Product Backlog

|    |      |               |                             |   |            |                                                                                                                                                                                                                                                                                                                                                                          |        |        |           |
|----|------|---------------|-----------------------------|---|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-----------|
| 26 | US24 | Priya/Meghana | Agent Decision Timeline     | F | 2026-03-11 | that I can understand how the system processed my dataset.<br><b>Acceptance Criteria</b><br>Each agent action is logged<br>Timeline shows sequence of agent operations<br>Each step displays agent name and action<br>Timeline available in final dataset report                                                                                                         | Small  | Medium |           |
| 27 | US25 | Shreya/Kush   | Dataset Profiling           | F | 2026-03-11 | provide summary statistics so that I can quickly understand the structure and distribution of my data.<br><b>Acceptance Criteria</b><br>System detects number of rows and columns<br>System identifies column data types<br>System calculates summary statistics for numeric columns<br>System identifies categorical column distributions<br>Profiling report generated | Medium | Medium |           |
| 28 | US26 | sprint 3      | Sandbox Dry-Run Mode        | F | 13-03-2026 | As a data engineer, I want the agent to support a sandbox dry-run mode on a small sample of the data so that I can see all planned transformations and health checks before committing changes to the full dataset.                                                                                                                                                      | Medium | Medium |           |
| 29 | US27 | sprint 3      | Auto-Generated Run Docs     | F | 13-03-2026 | As a product owner, I want the agent to generate human-readable documentation for each ETL run so that stakeholders can quickly understand what inputs, transformations, and outputs were involved.                                                                                                                                                                      | Small  | Medium | US26,US25 |
| 30 | US28 | sprint 3      | User Feedback & Rating Loop | F | 13-03-2026 | As an end user, I want to rate pipeline runs and leave feedback so that the agent can suggest improvements to mappings, validations, or configurations over time.                                                                                                                                                                                                        | Small  | Medium | US27      |

# Sprint 3 Backlog

| 1 | No.      | Description                                                                                                                                                | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Feature                   | Story points | Priority | Assignee           |
|---|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------|----------|--------------------|
| 2 | US_S2_01 | As a data steward, I want to validate and monitor data quality across the ETL pipeline so that errors are detected early and data integrity is maintained. | <ul style="list-style-type: none"><li>* ETL step execution logs are captured for each stage</li><li>* Row-level errors are detected and clearly reported</li><li>* Schema validation ensures input and output consistency</li><li>* Null values and anomalies are identified correctly</li><li>* Validation reports summarize all detected issues</li><li>* API returns validation results in structured format</li><li>* System prevents invalid data from progressing in pipeline</li></ul> | Data Validation & Quality | 55           | High     | Tenzing / Orges    |
| 3 | US_S2_02 | As a system, I want agents to coordinate and execute tasks in parallel so that ETL performance and scalability improve.                                    | <ul style="list-style-type: none"><li>* ETLState contains all required shared fields</li><li>* Agents execute tasks in correct workflow order</li><li>* Parallel execution runs without data conflicts</li><li>* Retry mechanism works correctly for failed steps</li><li>* Agent logs capture execution details and decisions</li><li>* Pipeline execution completes successfully under coordination</li><li>* System handles partial failures without crashing entire flow</li></ul>        | Multi-Agent System        | 55           | Medium   | Tenzing / Orges    |
| 4 | US_S2_03 | As a user, I want flexible data ingestion methods so that I can easily upload datasets from multiple sources.                                              | <ul style="list-style-type: none"><li>* File upload supports drag &amp; drop functionality</li><li>* API-based upload accepts valid file formats (CSV, JSON)</li><li>* Upload progress is displayed to the user</li><li>* Invalid files are rejected with clear error messages</li><li>* Uploaded datasets are stored successfully</li><li>* System handles large file uploads without failure</li></ul>                                                                                      | Data Input                | 34           | High     | Shreya / Kushawnth |
| 5 | US_S2_04 | As a user, I want to export processed data in multiple formats so that I can use it externally.                                                            | <ul style="list-style-type: none"><li>* Processed data can be exported in CSV format</li><li>* Processed data can be exported in JSON format</li><li>* Processed data can be exported in Excel format</li><li>* Download API returns correct file response</li><li>* Exported data maintains correct schema and formatting</li><li>* System handles export errors gracefully</li></ul>                                                                                                        | Data Output               | 34           | High     | Tenzing / Orges    |

# Sprint 3 Backlog

|   |          |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                             |                    |    |        |                    |
|---|----------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----|--------|--------------------|
| 6 | US_S2_05 | As a user, I want to preview data before and after processing so that I understand transformations clearly. | <ul style="list-style-type: none"><li>* Dataset preview displays tabular data correctly</li><li>* Schema view shows column names and data types</li><li>* ERD visualization represents dataset relationships</li><li>* Before and after data comparison is available</li><li>* UI loads data efficiently without performance issues</li><li>* API returns preview data accurately</li></ul> | Data Visualization | 34 | Medium | Shreya / Kushawnth |
| 7 | US_S2_06 | As a user, I want basic analytics and filtering tools so that I can quickly analyze datasets.               | <ul style="list-style-type: none"><li>* System calculates average and median correctly</li><li>* Summary statistics are generated for datasets</li><li>* Backend supports filtering queries</li><li>* Aggregation results are returned in structured format</li><li>* Queries execute efficiently on large datasets</li><li>* Errors are handled and reported clearly</li></ul>             | Analytics          | 34 | Medium | Tenzing / Orges    |
| 8 | US_S2_07 | As a compliance officer, I want sensitive data to be redacted so that user privacy is protected.            | <ul style="list-style-type: none"><li>* Schema changes are detected between pipeline runs</li><li>* System identifies added, removed, or modified fields</li><li>* Alerts are generated for unexpected schema changes</li><li>* Schema change logs are stored for auditing</li><li>* Monitoring integrates with ETL pipeline execution</li></ul>                                            | Security           | 21 | Medium | Tenzing / Orges    |



# Sprint 3 Stories Completed

| 1 | No.      | Description                                                                                                                                                                        | Acceptance Criteria                                                                                                                                                                                                                                                                              | Priority | Feature                            | Story point | Assignee    |
|---|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------|-------------|-------------|
| 2 | US_S3_01 | As a data engineer, I want the ETL pipeline to complete extraction and transformation within a reasonable time threshold so that the system is practical for real-world workloads. | <ul style="list-style-type: none"><li>* Pipeline completes within defined time threshold</li><li>* Performance improves compared to previous runs</li><li>* Bottlenecks are identified and optimized</li><li>* Execution metrics are logged</li><li>* Data accuracy remains unaffected</li></ul> | High     | Pipeline Reliability & Performance | 11          | Orges Velia |
| 3 | US_S3_02 | As a compliance officer, I want API keys and database credentials to be stored securely so that sensitive data is not exposed.                                                     | <ul style="list-style-type: none"><li>* Credentials are not hardcoded</li><li>* Sensitive data is not logged</li><li>* Secure storage mechanism is used</li><li>* Access is restricted properly</li><li>* Data exposure risks are minimized</li></ul>                                            | High     | Security & Resource Optimization   | 5           | Orges Velia |
| 4 | US_S3_03 | As a developer, I want the pipeline to be modular and independently testable so that changes can be made without affecting the entire system.                                      | <ul style="list-style-type: none"><li>* Pipeline components are modular</li><li>* Changes do not break other modules</li><li>* Code is reusable and clean</li><li>* New features can be added easily</li><li>* Documentation is updated</li></ul>                                                | High     | Security & Resource Optimization   | 8           | Orges Velia |
| 5 | US_S3_04 | As a data steward, I want every agent action to be logged with execution details so that pipeline runs can be audited.                                                             | <ul style="list-style-type: none"><li>* All actions are logged with timestamps</li><li>* Input/output summaries are stored</li><li>* Logs are retrievable</li><li>* Audit trail is complete</li><li>* Logs are structured and readable</li></ul>                                                 | High     | Traceability & Audit Logging       | 8           | Kush        |

# Sprint 3 Stories Completed

| No. | Description                                                                                                        | Acceptance Criteria                                                                                                                                                                                                                         | Priority | Feature                        | Story point | Assignee         |
|-----|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------|-------------|------------------|
| 7   | US_S3_06<br>As a user, I want to validate dataset quality so that I can detect issues before processing.           | <ul style="list-style-type: none"><li>* Missing values are detected</li><li>* Duplicate records are identified</li><li>* Data types are validated</li><li>* Validation summary is generated</li><li>* Errors are clearly reported</li></ul> | Medium   | Data Quality & Health Insights | 8           | Priya Prajapati  |
| 8   | US_S3_07<br>As a user, I want to rate pipeline runs and provide feedback so that the system can improve over time. | <ul style="list-style-type: none"><li>* Users can submit ratings</li><li>* Feedback is stored successfully</li><li>* Comments can be added</li><li>* Feedback is linked to runs</li><li>* UI is easy to use</li></ul>                       | Medium   | User Feedback & Learning Loop  | 5           | Shreya Kuvalekar |
| 9   | US_S3_08<br>As a user, I want to connect a database to the system so that I can ingest structured data.            | <ul style="list-style-type: none"><li>* Database connection is successful</li><li>* Data can be fetched correctly</li><li>* Connection errors are handled</li><li>* Data is validated</li><li>* Logs capture interaction</li></ul>          | High     | Data Source Integration        | 8           | Orges Velia      |
| 10  | US_S3_09<br>As a user, I want to connect API-based datasets so that I can ingest external data into the system.    | <ul style="list-style-type: none"><li>* API connections work correctly</li><li>* Data is parsed successfully</li><li>* Errors are handled properly</li><li>* Data validation is applied</li><li>* API interactions are logged</li></ul>     | Medium   | Data Source Integration        | 8           | Tenzing Salaka   |

# Sprint 3 Test cases

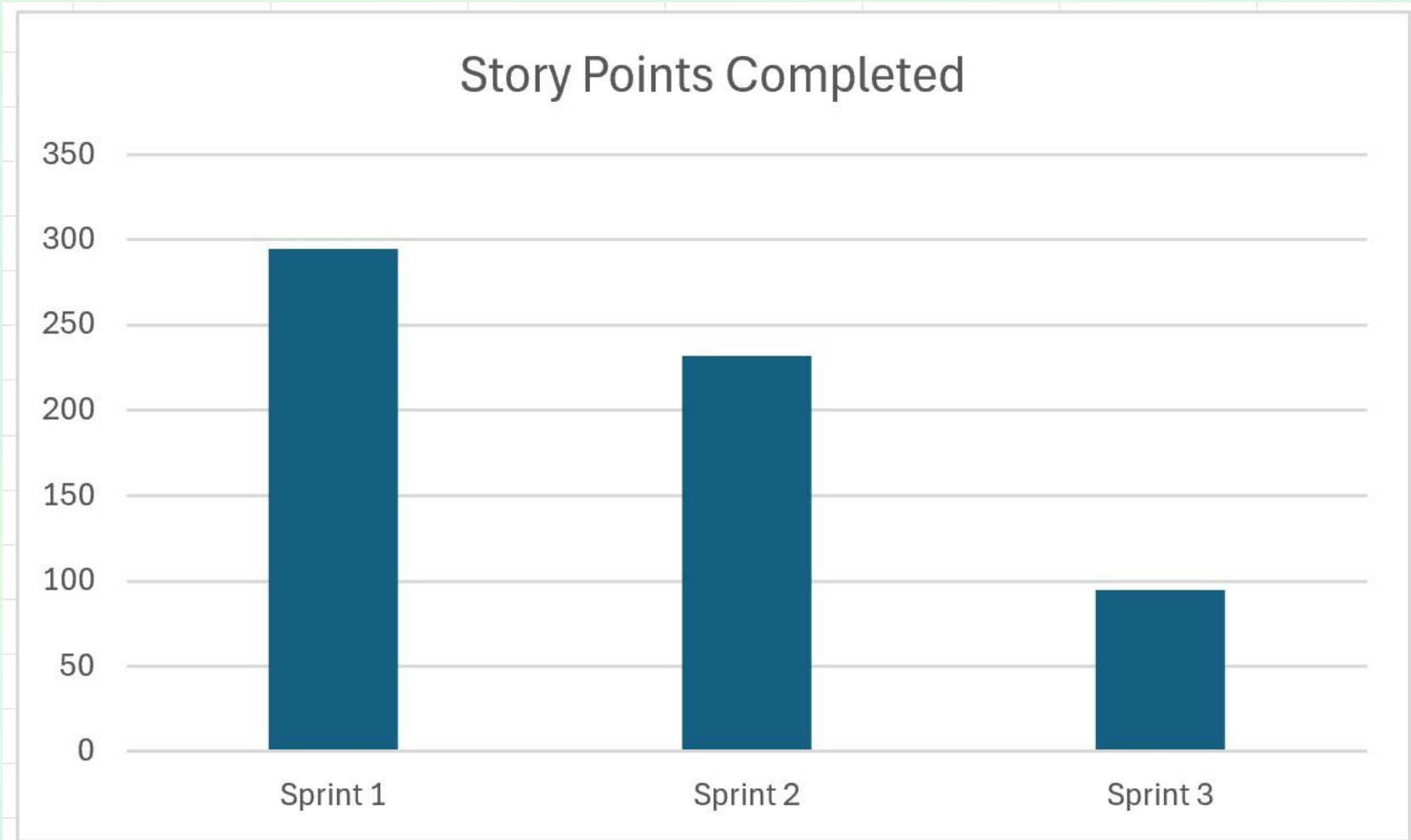
| 1  | TC ID    | Module                             | Scenario                                | Priority | Status |
|----|----------|------------------------------------|-----------------------------------------|----------|--------|
| 2  | TC_S3_01 | Pipeline Reliability & Performance | ETL pipeline execution within threshold | High     | Pass   |
| 3  | TC_S3_02 | Pipeline Reliability & Performance | Execution metrics logging               | High     | Pass   |
| 4  | TC_S3_03 | Pipeline Reliability & Performance | Bottleneck detection                    | High     | Pass   |
| 5  | TC_S3_04 | Pipeline Reliability & Performance | Data accuracy after optimization        | High     | Pass   |
| 6  | TC_S3_05 | Security & Resource Optimization   | No hardcoded credentials                | High     | Pass   |
| 7  | TC_S3_06 | Security & Resource Optimization   | Sensitive info not logged               | High     | Pass   |
| 8  | TC_S3_07 | Security & Resource Optimization   | Secure credential storage               | High     | Pass   |
| 9  | TC_S3_08 | Security & Resource Optimization   | Restricted credential access            | High     | Pass   |
| 10 | TC_S3_09 | Security & Resource Optimization   | Modular pipeline components             | High     | Pass   |
| 11 | TC_S3_10 | Security & Resource Optimization   | No cross-module impact                  | High     | Pass   |
| 12 | TC_S3_13 | Traceability & Audit Logging       | Agent actions logging                   | High     | Pass   |



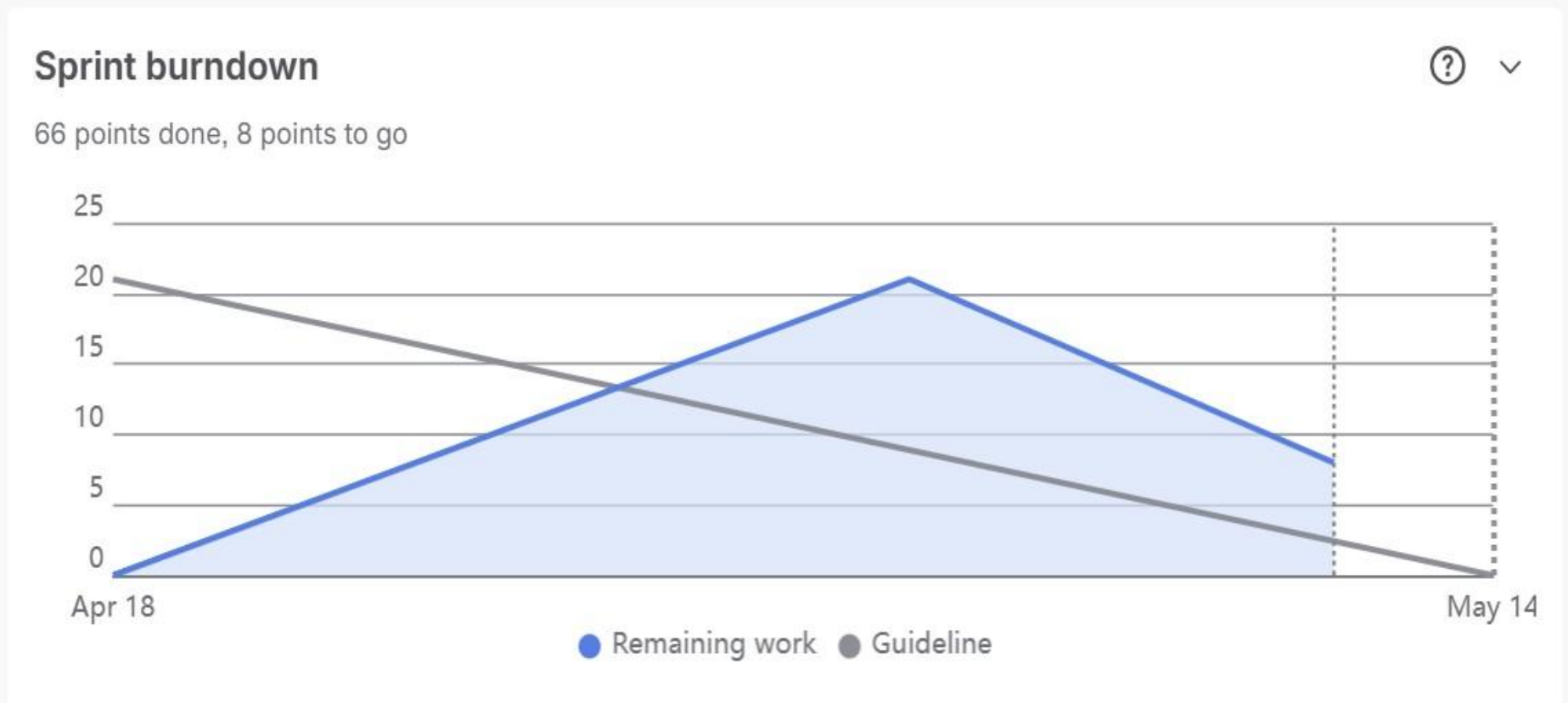
# Sprint 3 Test cases

| 1  | TC ID    | Module                             | Scenario                          | Priority | Status |
|----|----------|------------------------------------|-----------------------------------|----------|--------|
| 13 | TC_S3_14 | Traceability & Audit Logging       | Input/output summaries stored     | High     | Pass   |
| 14 | TC_S3_15 | Traceability & Audit Logging       | Log retrieval                     | High     | Pass   |
| 15 | TC_S3_16 | Traceability & Audit Logging       | Audit trail completeness          | High     | Pass   |
| 16 | TC_S3_18 | Pipeline Reliability & Performance | Large dataset handling            | High     | Pass   |
| 17 | TC_S3_19 | Pipeline Reliability & Performance | Performance under load            | High     | Pass   |
| 18 | TC_S3_20 | Pipeline Reliability & Performance | No crashes under heavy processing | High     | Pass   |
| 19 | TC_S3_31 | Data Source Integration            | Database connection success       | High     | Pass   |
| 20 | TC_S3_32 | Data Source Integration            | Structured data fetching          | High     | Pass   |
| 21 | TC_S3_33 | Data Source Integration            | DB error handling                 | High     | Pass   |

# Team Velocity



# Burndown Chart





# Completed/Committed Ratio



# Sprint 3 Retrospective

- **What went well**

- All committed work was completed (**100% completion**)
- Strong delivery in **pipeline performance, security, and audit logging**
- System stability improved with better handling of workloads and integrations

- **What to improve**

- **Velocity dropped** compared to earlier sprints
- Less focus on **medium-priority features** like data quality and user feedback
- Need better **workload distribution and planning accuracy**

- **Action items**

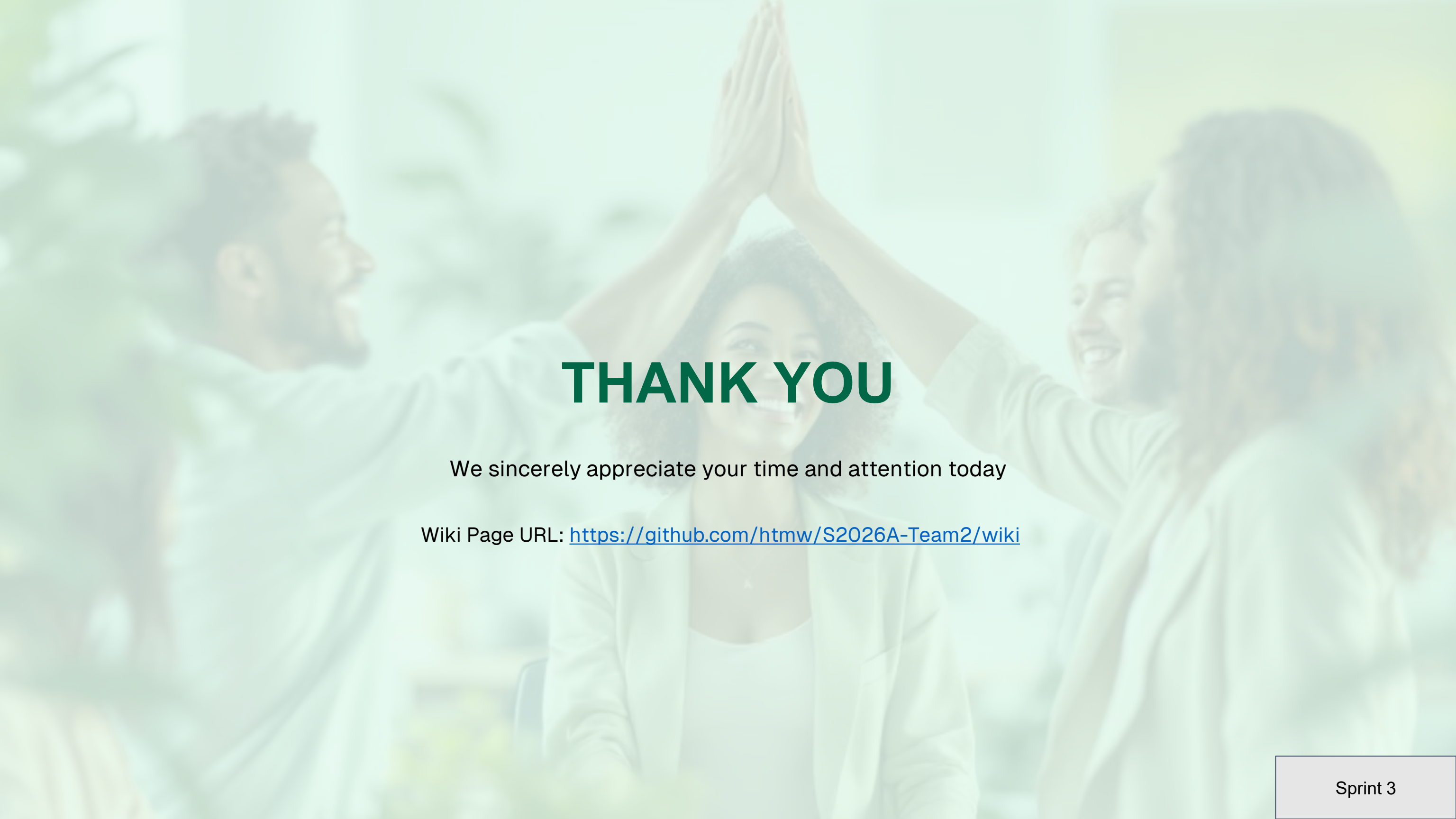
- Improve **estimation and sprint planning**
- Balance focus between **core and enhancement features**
- Increase **testing, especially for scalability**
- Enhance **documentation and user-facing features**



# Sprint 3 Demo

Demo [Video](#)



A background image showing three people (two men and one woman) smiling and high-fiving each other. The image is faded and serves as a backdrop for the text.

# THANK YOU

We sincerely appreciate your time and attention today

Wiki Page URL: <https://github.com/htmw/S2026A-Team2/wiki>